Optimization and Curve Sketching

Critical points and the first/second derivative tests

1 Find the critical points of $f(x) = x^3 - 6x^2 + 9x$, and classify each as a local max or min using the second derivative test.

2 For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.

Optimization applications

3 A farmer has 200 m of fencing to enclose a rectangular field against an existing wall (so only 3 sides need fencing). Find the dimensions that maximize the enclosed area.

4 An open-top box is made from a 12 cm square sheet by cutting equal squares of side x from each corner and folding up the sides. Find the value of x that maximizes the box's volume.

More optimization

5 A cylindrical can must have volume 500π cm^3. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi rh$, find the radius that minimizes the material used (surface area).

6 Find two positive numbers whose sum is 20 and whose product is as large as possible.

Curve sketching

7 For $f(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$, find the intervals of increase and decrease.

8 Using the results above, classify $x = -1$ and $x = 3$ as local maxima or minima via the first derivative test.

Minimizing distance and cost

9 Find the point on the parabola $y = x^2$ closest to the point $(0, 3)$, by minimizing $D^2 = x^2 + (x^2 - 3)^2$.

10 A rectangular box with a square base and open top must have volume 32 m^3. Find the base side length that minimizes the surface area, using $S = x^2 + \frac{128}{x}$ (derived from $V = x^2 h = 32$).

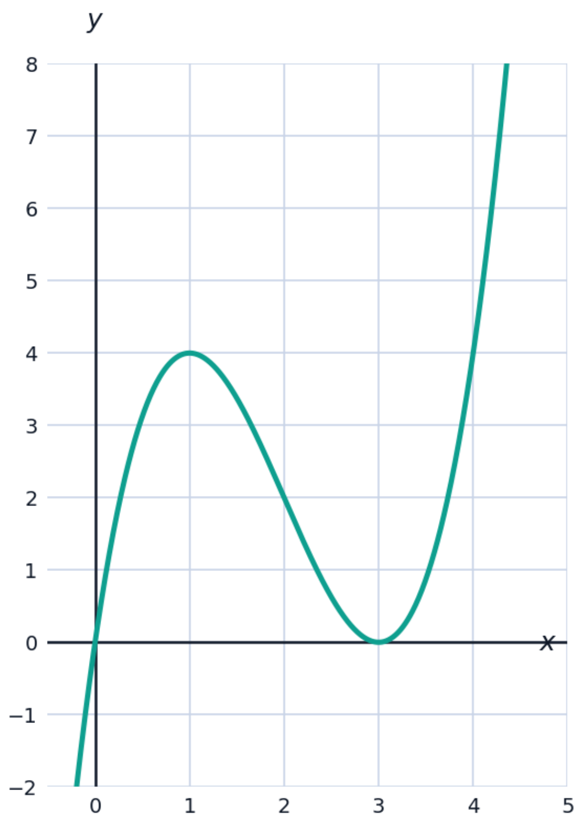
Sketching using derivative information

11 For $f(x) = 3x^4 - 4x^3$, find all critical points and classify each using the second derivative test.

12 For $f(x) = xe^{-x}$, find the critical point and determine whether it is a maximum or minimum.

Visualizing critical points on a graph

13 The graph shows $f(x) = x^3 - 6x^2 + 9x$ from the first problem. Use the graph to confirm the local max at $x = 1$ and local min at $x = 3$ found algebraically.



Optimization with a fixed perimeter

14 A rectangular garden must have a perimeter of 80 m. Find the dimensions that maximize its area.

15 A Norman window (a rectangle topped by a semicircle) has perimeter 10 m, with rectangle width x and height y: $2y + x + \pi\frac{x}{2} = 10$. Express the total area $A = xy + \frac{\pi}{8}x^2$ in terms of x alone, and state the equation you would set $A'(x) = 0$ to solve (do not solve).

Concavity and the second derivative test in context

16 A company's profit is modeled by $P(x) = -2x^2 + 80x - 300$, where x is units sold (hundreds). Find the production level that maximizes profit, and verify it is a maximum using the second derivative.